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ΘΕΣΣΑΛΟΝΙΚΗΣ

Signal Integrity on 2.5D Integrated Systems

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- Silicon Interposers
- Signal Integrity assessment
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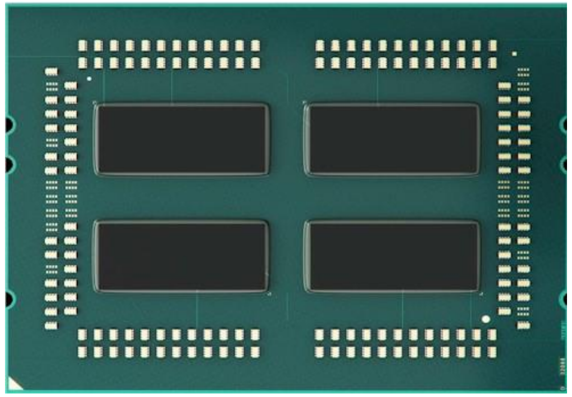
Introduction – Motivation

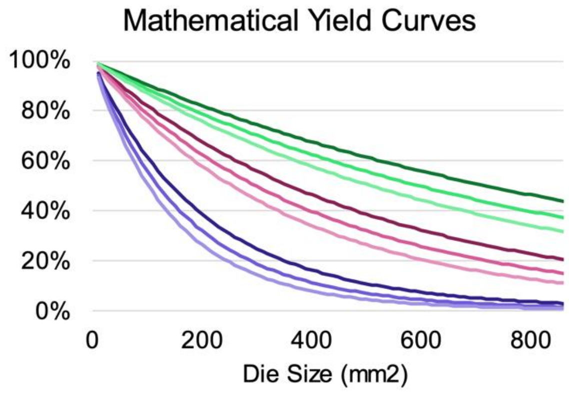


AMD EPYC 7000

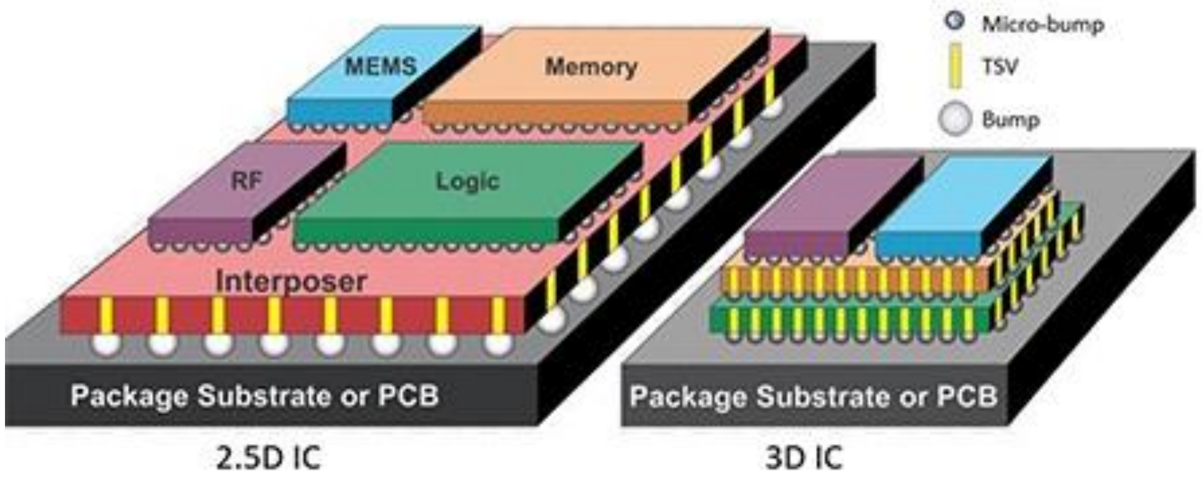
High Demand for electronics with:

- Low Cost
- Low Latency
- High Computational Power

- Semiconductor manufacturing yield decreases nonlinearly as a function of the die size
- Manufacturing defects are a major problem



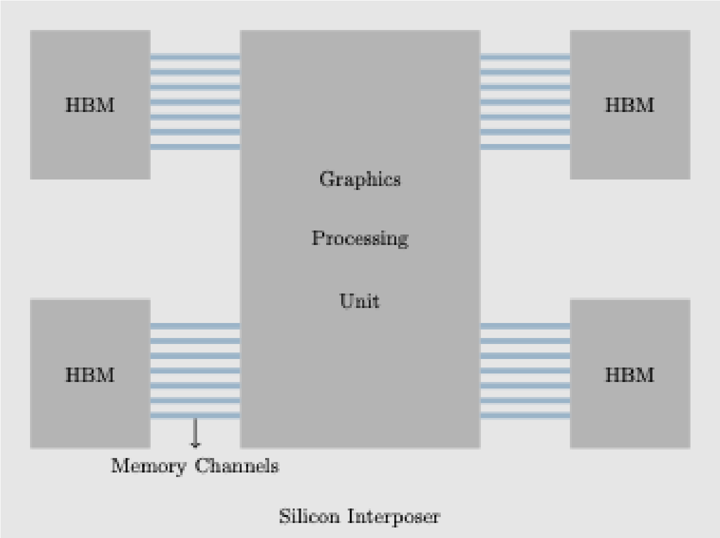
Introduction – Motivation

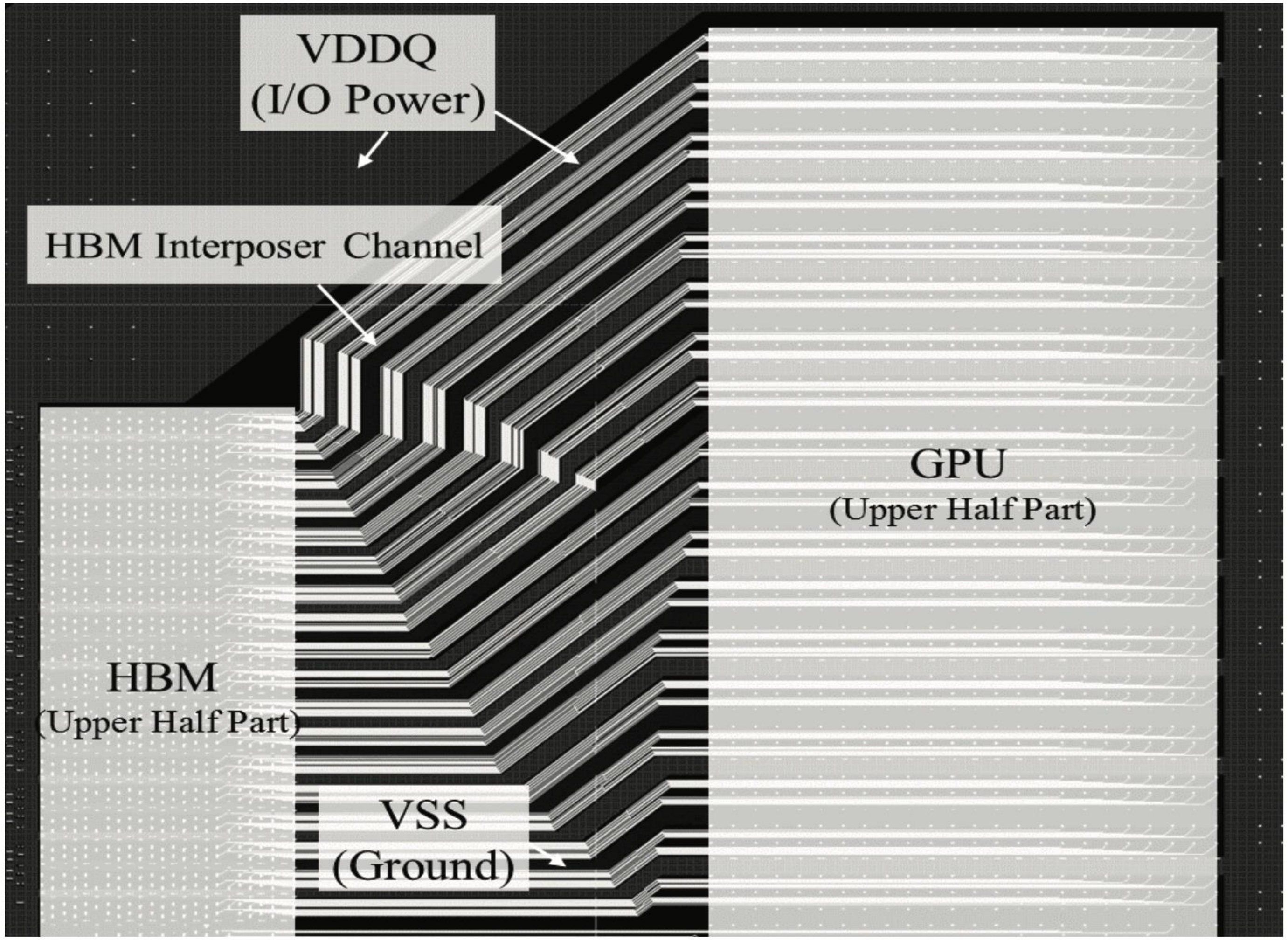


- Silicon Interposers are an advanced packaging solution
- Vehicle / bridge that connects all the different ICs
- Provide a high-density routing layer
  - Signal
  - Power
- Exploits μ – Bump and TSV technologies

Silicon Interposers

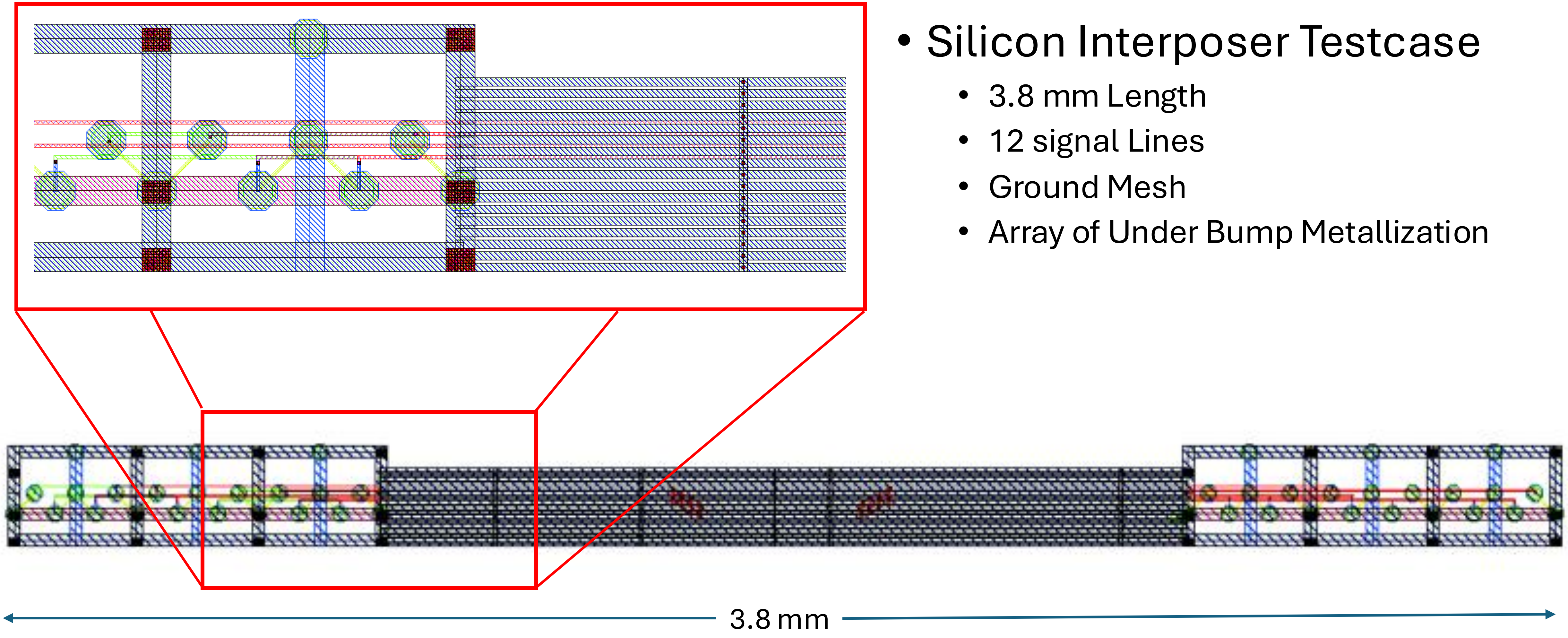
- Real link between a HBM module and a GPU
  - Thousands of interconnections
  - Mostly parallel signal lines
  - Divided into groups
  - Coupling (Capacitive and Inductive) between the lines





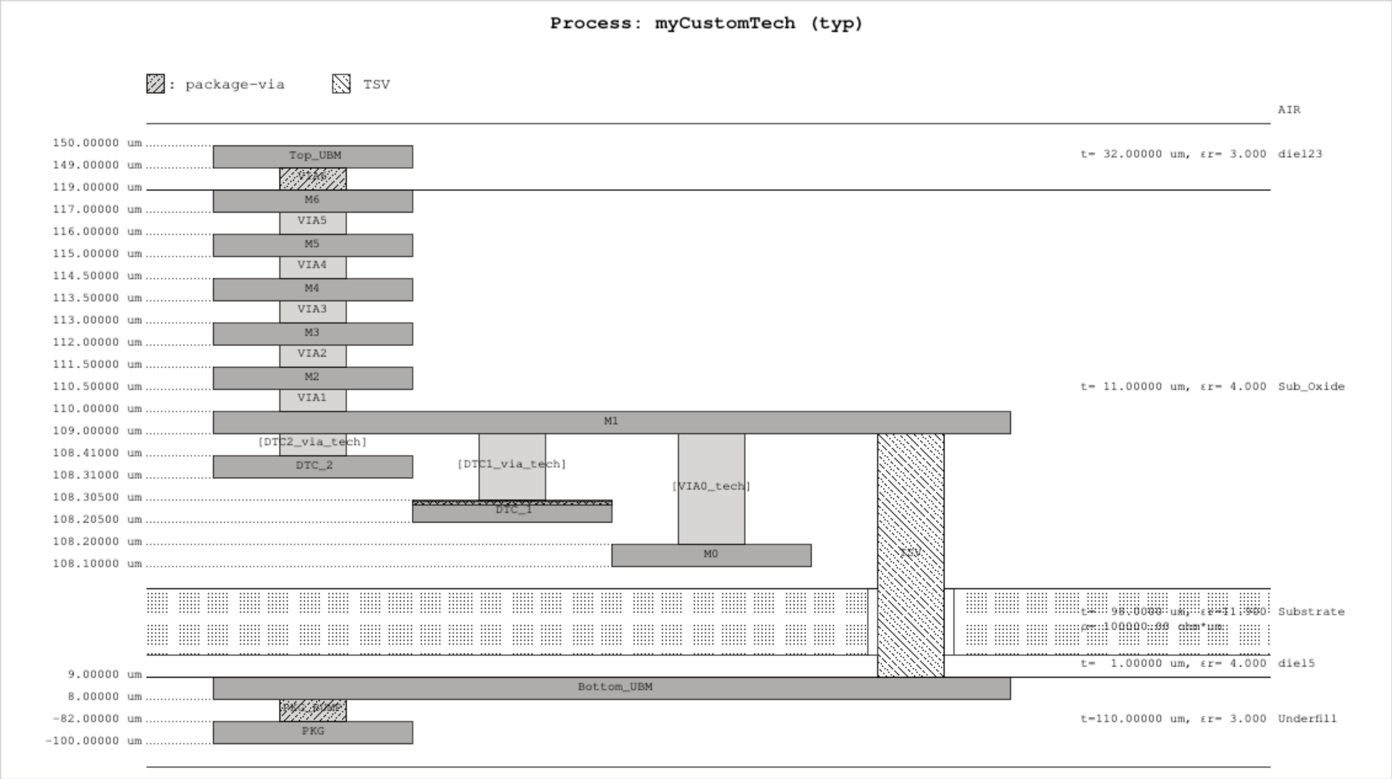
Silicon Interposers

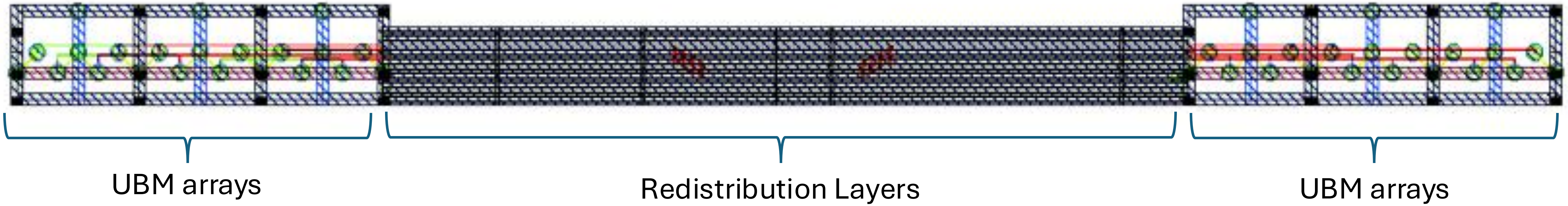
- Silicon Interposer Testcase
  - 3.8 mm Length
  - 12 signal Lines
  - Ground Mesh
  - Array of Under Bump Metallization



Silicon Interposers

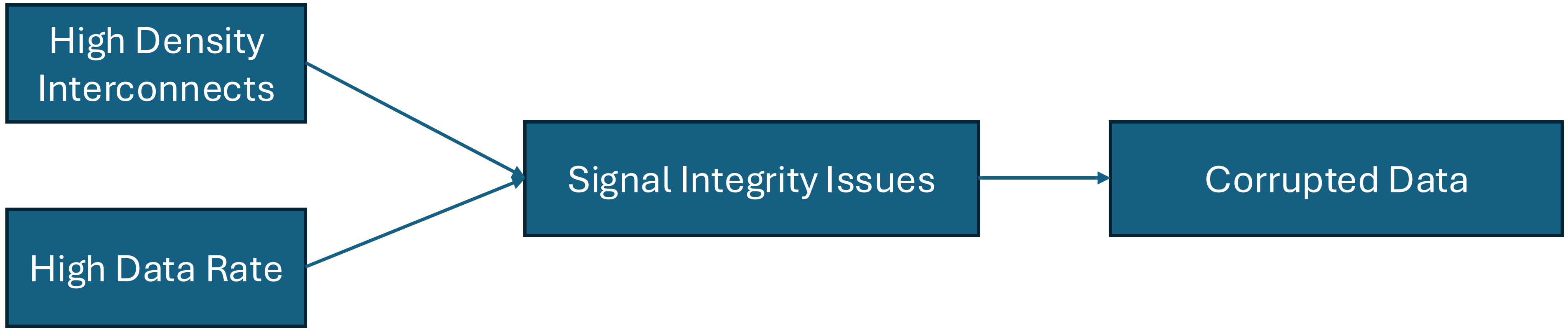
- Silicon Interposer Testcase
- 6 Metals
  - M1, M3, M5 Signal lines
  - M2, M4, M6 Ground Mesh
- The Ground Mesh satisfies both
  - Structural
  - Electrical Constraints





Signal Integrity assessment

- Silicon Interposers are large structures
- Require **EM simulations** for verification and optimization prior to manufacturing
- These simulations are **time consuming** with **high computational cost**



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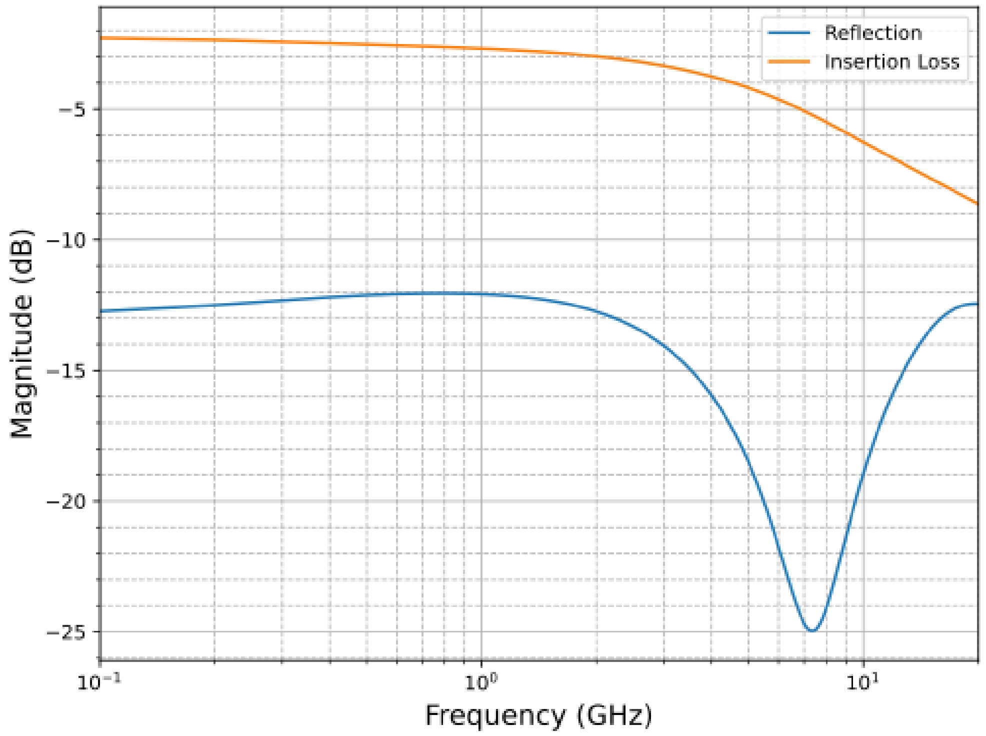
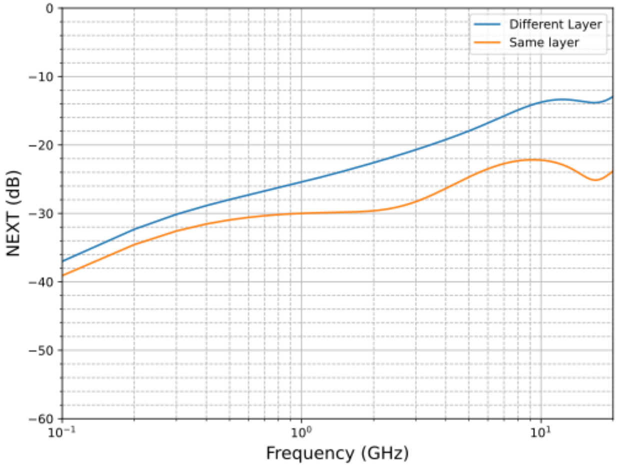
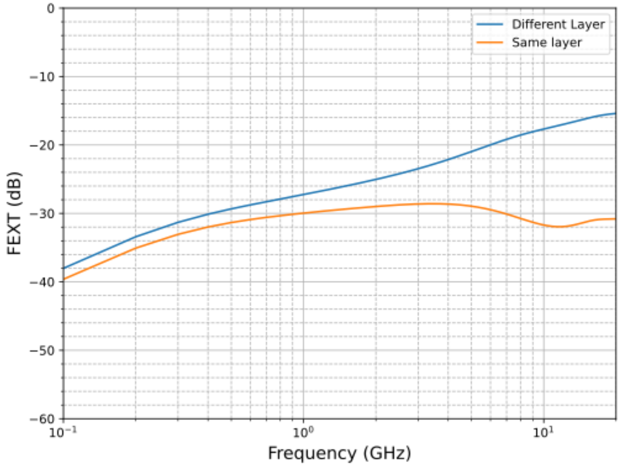
Signal Integrity assessment

- EM Simulations
- The geometry is discretized into small elements
- Maxwells equations are solved for each element
- Results: **S – parameters** that describe the electrical characteristics of the structure (Frequency Domain)



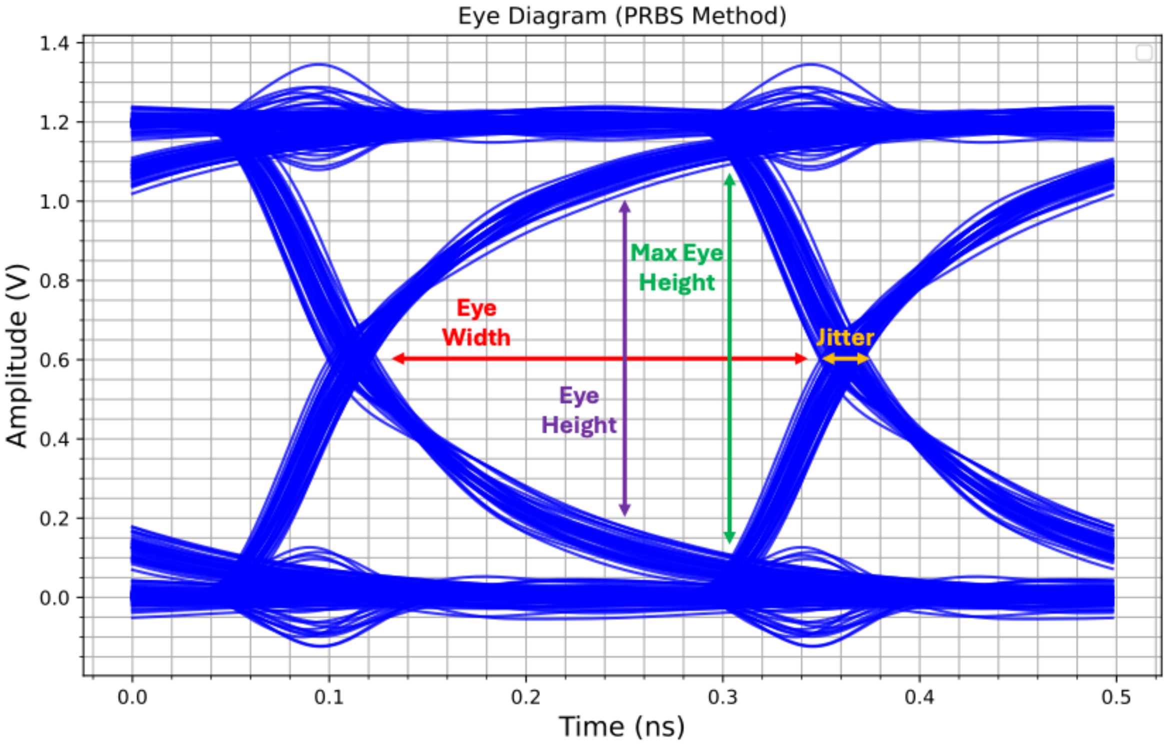




Signal Integrity assessment

- SPICE Simulators can handle S – parameters by converting them into equivalent circuit models
- Time domain simulations can be executed (**Eye Diagrams**)

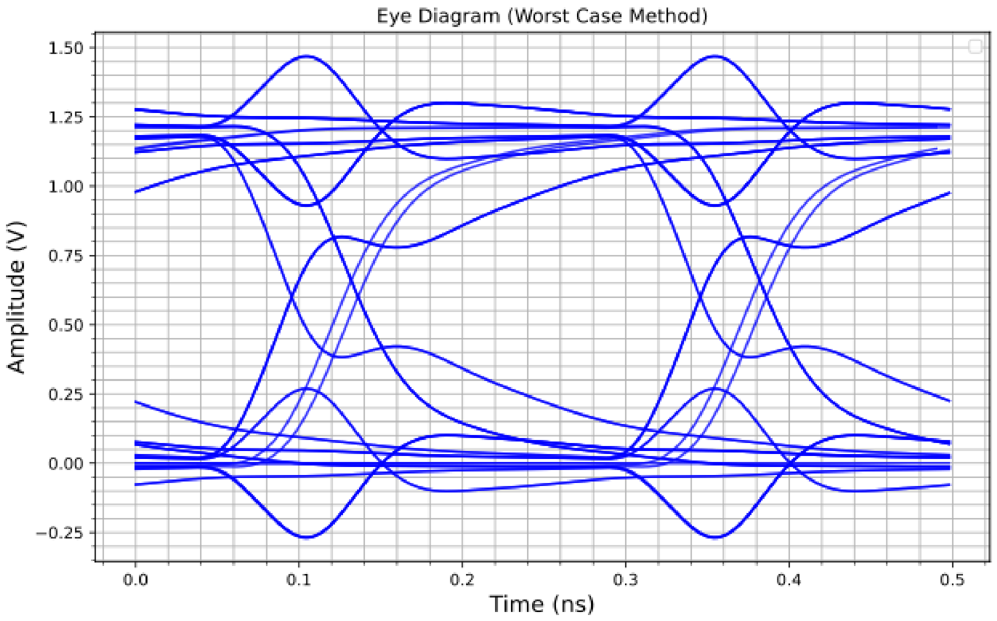
- They can be equally time consuming, usually neglected from bibliography
- Need for PRBS with 2^8 – 2^21 bits for high accuracy
- Crosstalk, ringing, mismatches can be investigated



Signal Integrity assessment

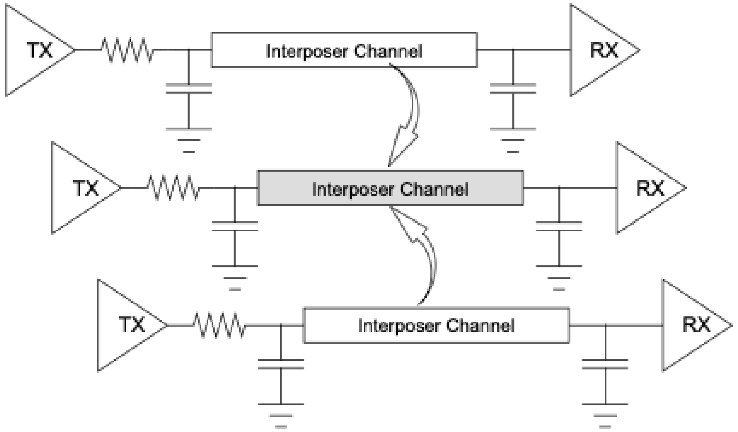
- To accelerate Eye Diagram Extraction a **Worst-Case Method** was implemented
- The line of interest is considered as a **victim line**
- All the neighbor lines are the **aggressors**
- The aggressors go through equal phase transitions leading to **constructive interference**
- Eye Diagrams are extracted in fewer periods (8 w.c. transitions)
- Total acceleration of **37**× times

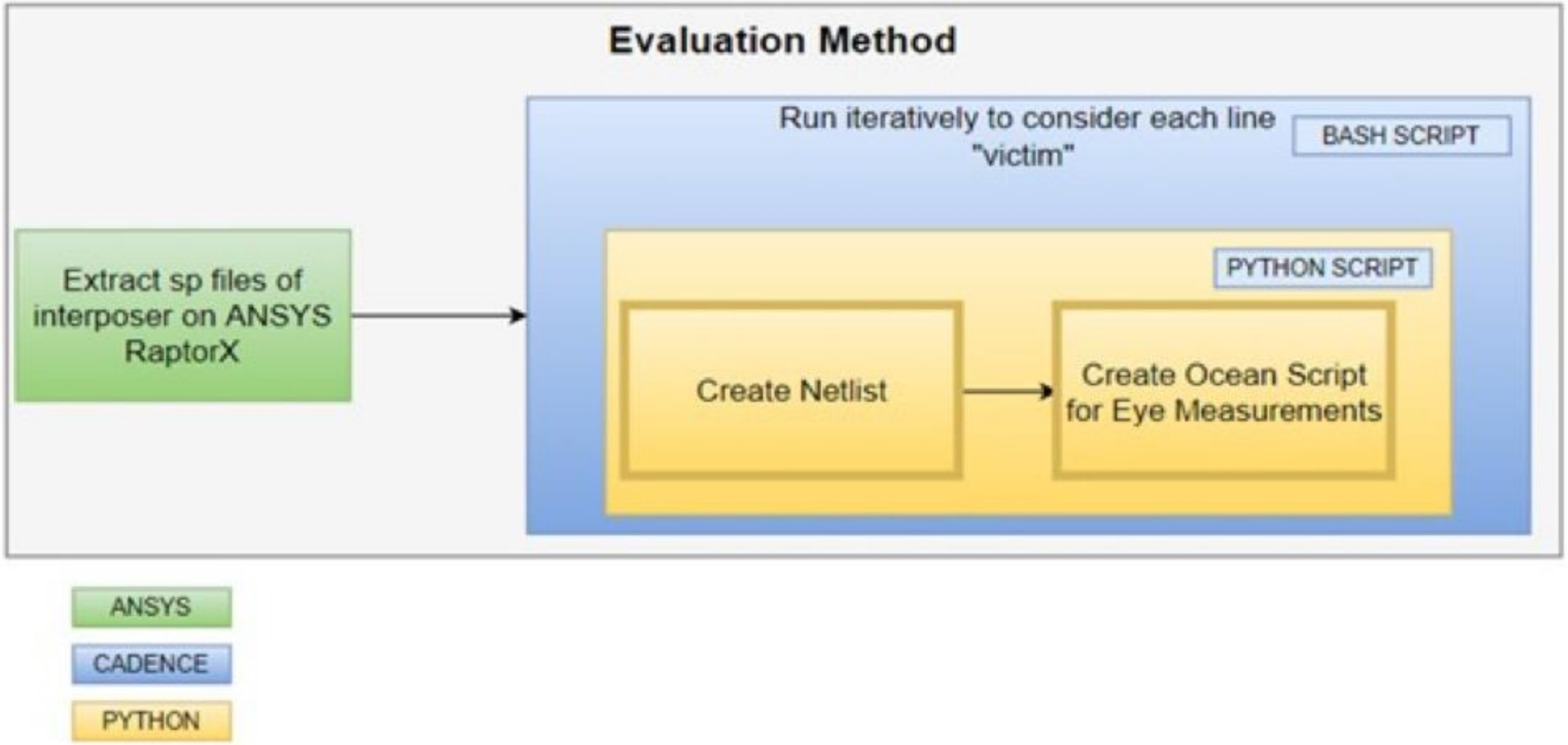
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Boundary line	Rise_Pos	Rise_Neg	Fall_Pos	Fall_Neg	V_H_Pos	V_H_Neg	V_L_Pos	V_L_Neg
Agg1								
Vic								
Agg2								



Signal Integrity assessment

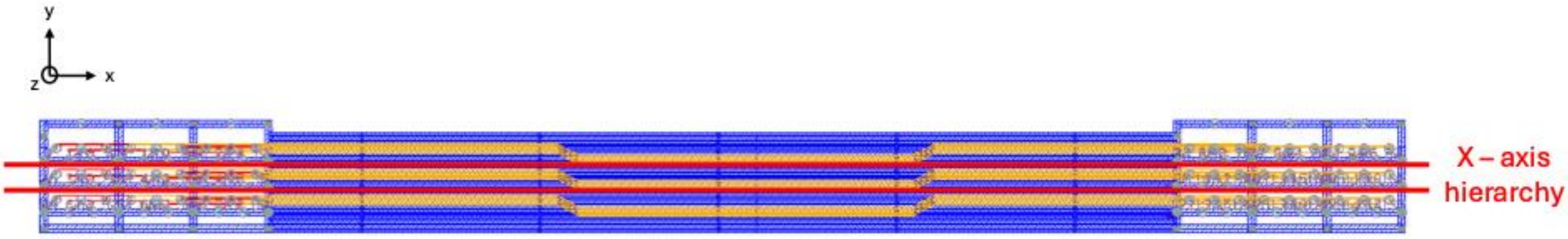
- Final Evaluation Method





Hierarchical Approaches

- To accelerate EM Simulations a hierarchical partition can be applied
- The structure can be partitioned into separate groups (X – axis) to lighten up the simulator

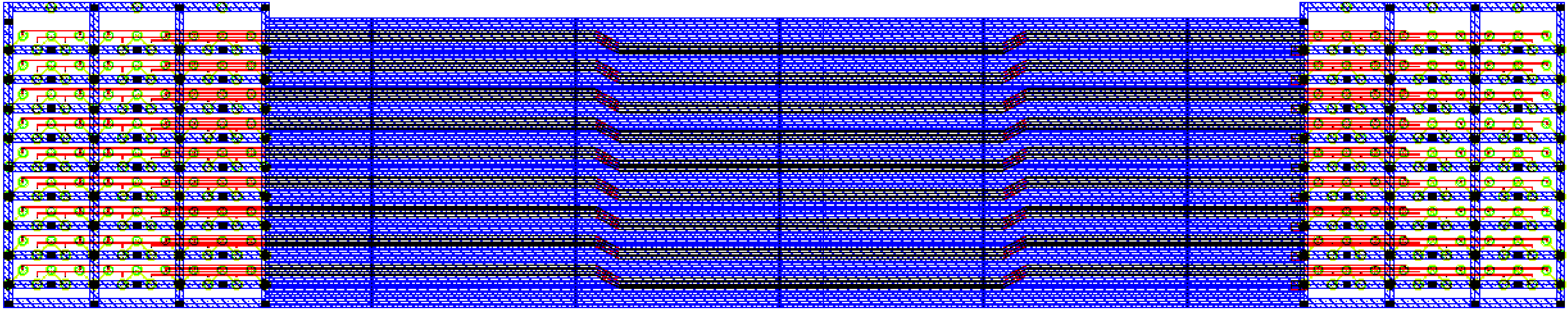


3 group interposer

Hierarchical Approaches

X – axis partition

- A multi group interposer can be approached with fewer groups
- Ignore distant groups, since coupling is reduced with distance



9 group interposer

Hierarchical Approaches

X – axis partitioning

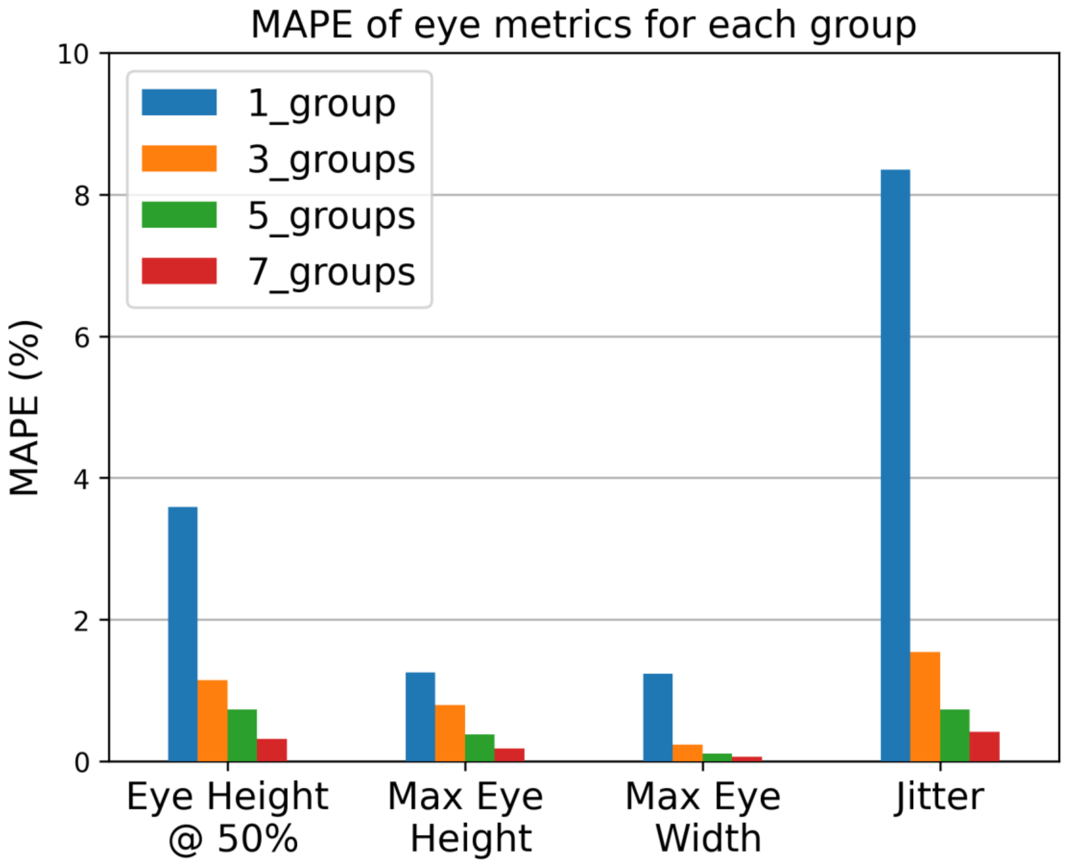
Test-case	RaptorX - Runtime	RaptorX - Memory	Spectre - Runtime	Speed-up	Memory Improvement
1 Group	5m:5.6s	10.786 GB	3m 56.9s	X 120.99	97.4%
3 Groups	33m:8.4s	57.698 GB	35m 22.4s	X 18.59	86.0%
5 Groups	3h:5m:50.5s	140.313 GB	1h 52m 49s	X 3.32	65.9%
7 Groups	6h:4m:1.3s	258.213 GB	3h 23m 13s	X 1.69	37.2%
9 Groups	10h:16m:13.9s	411.332 GB	5h 27m 29s	-	-

Speed-up and Memory improvement

Hierarchical Approaches

X – axis partitioning

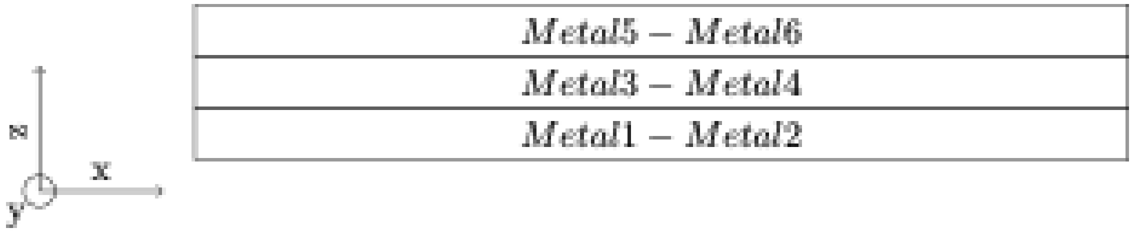
- A 9 – group interposer can be approached by 1 – group with:
- < 8% MAPE
- 120× acceleration
- 97 % Memory improvement

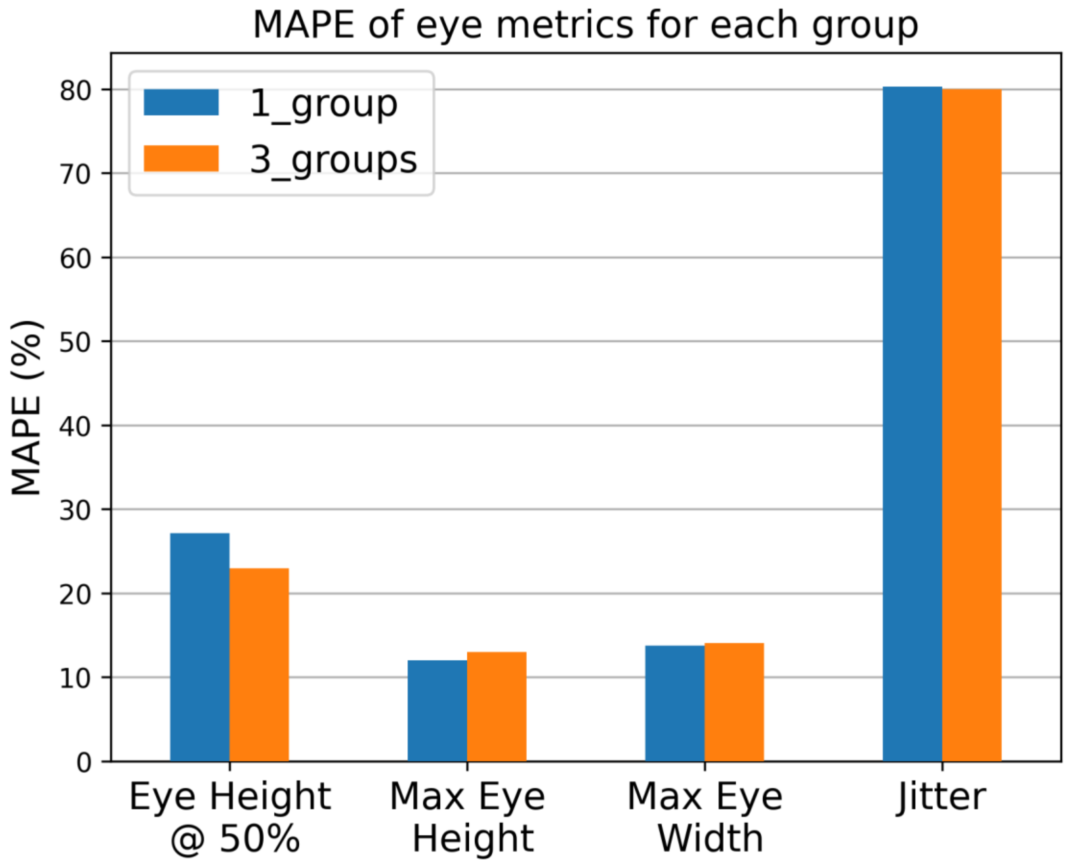


Hierarchical Approaches

Z – axis partitioning

- Intraplane coupling is dominant
- Important information is lost during Z – axis partitioning
- No major acceleration and memory improvement





Machine Learning Approaches

- Signal Integrity assessment of silicon interposers can be also accelerated with **Machine Learning** Techniques
- It is a **Regression Problem** with **high computational cost** and **high dimensional dataset**

Goal:

- Fast Signal Integrity analysis $\Longrightarrow$ Eye Diagram Metrics Prediction

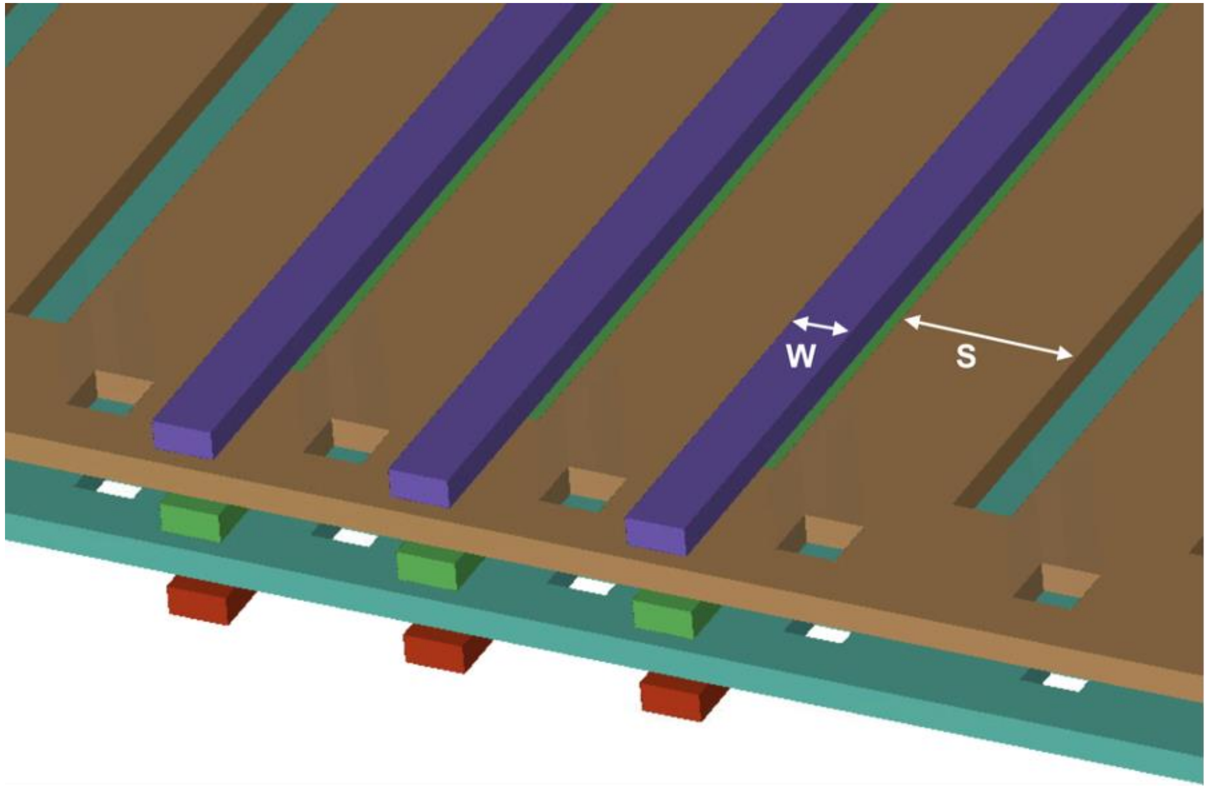
Challenges:

- Dataset Extraction

Machine Learning Approaches

Input Parameters	
Width	Geometry Parameters
Space	
Length	
Metal Thickness	PDK Parameters
Dielectric Thickness	
Signal Voltage	Communication Protocol Parameters
Data rate	
Rise Time	
Transmitter Resistance	

- Dataset Extraction for a simple RDL
- Ansys HFSS, 600 points



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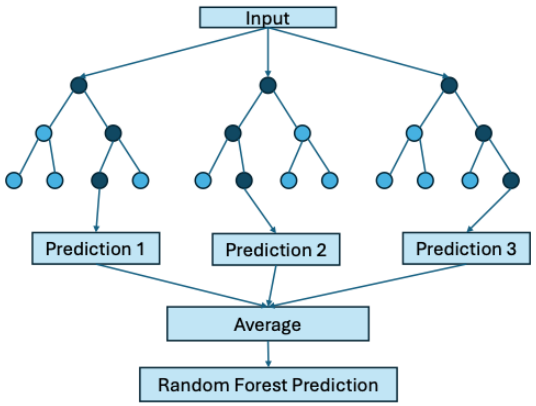
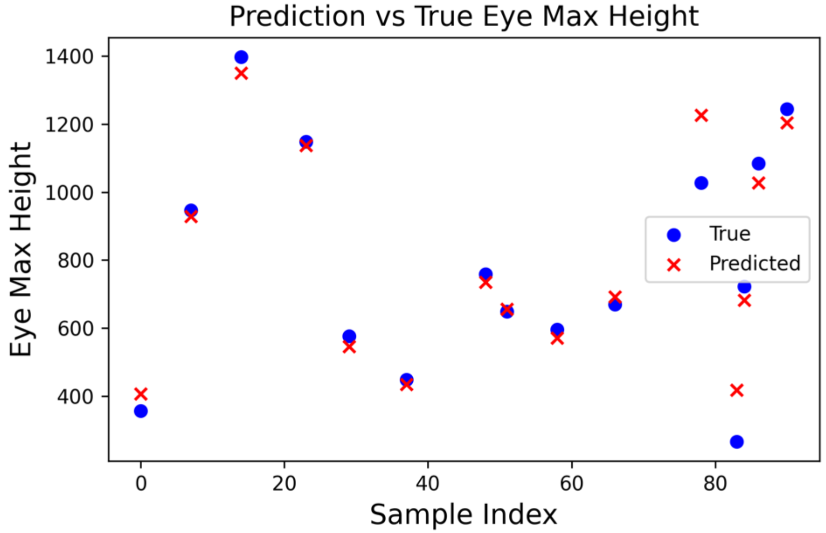
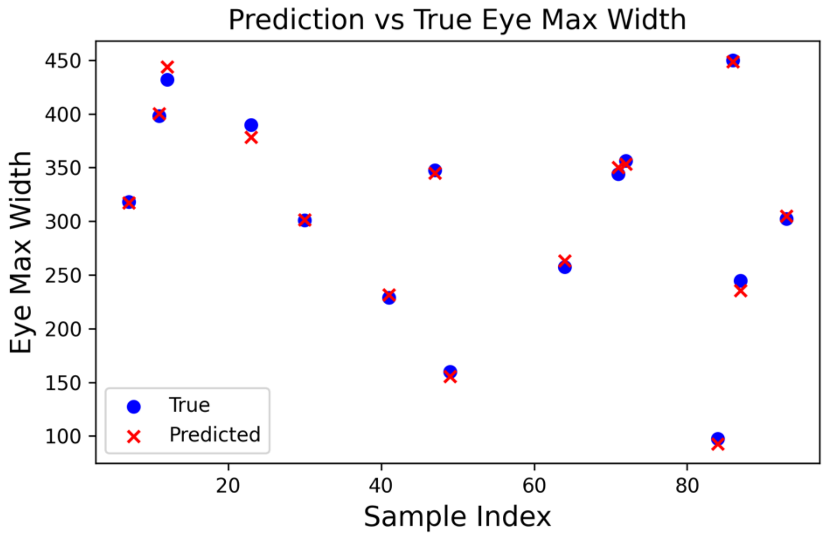
Machine Learning Approaches

- Random Forest Regression



Metrics	MAPE
Eye Max Height	10.90 %
Eye Max Width	3.34 %
Eye Height @ 50 % of UI	13.50 %

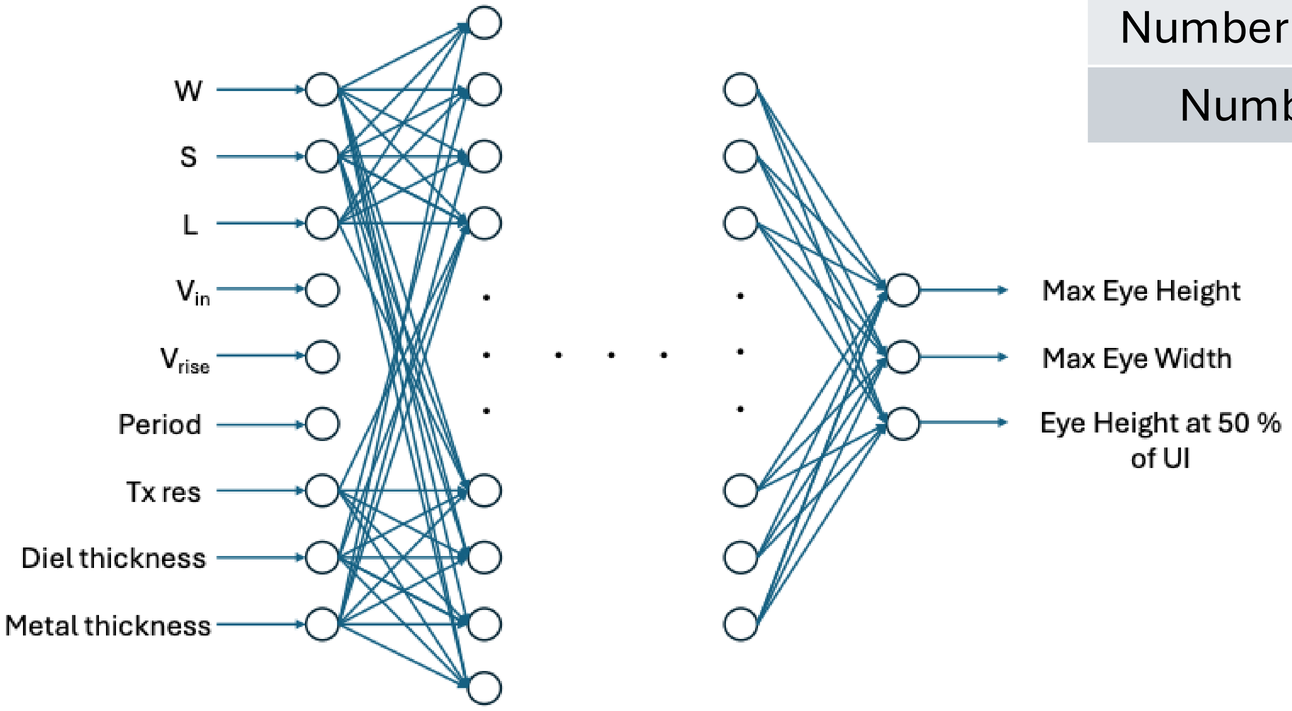




Machine Learning Approaches

- Deep Neural Network

Parameter	Value
Learning Rate	0.0005
Number of Epochs	70
Batch Size	8
Number of Hidden Layers	5
Number of Neurons	64, 128, 128, 128, 64

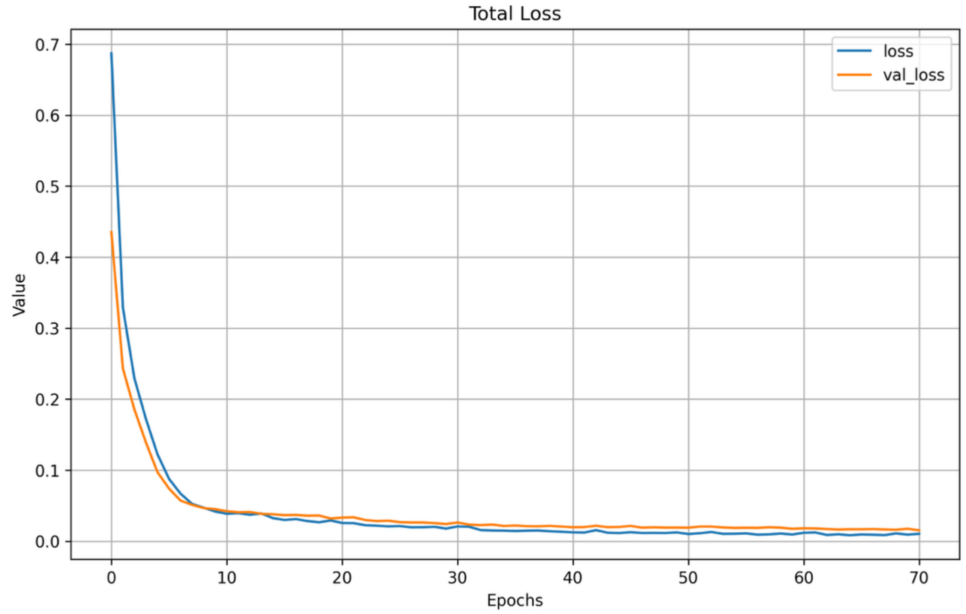


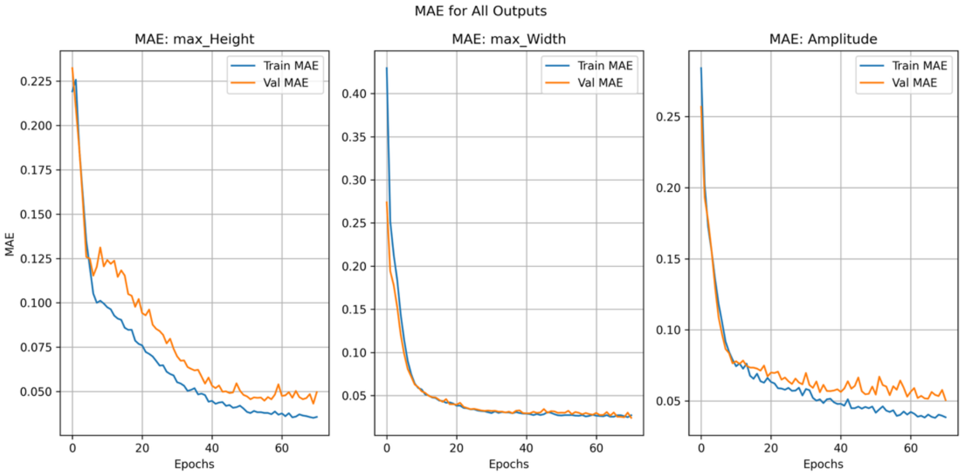
Machine Learning Approaches

Deep Neural Network

- It can achieve better accuracy
- Better cooperation with the highly complex and nonlinear relationships of the dataset
- Early – stop mechanism is exploited to stem overfitting

Metrics	MAPE
Eye Max Height	4.15 %
Eye Max Width	2.94 %
Eye Height @ 50 % of UI	5.08 %

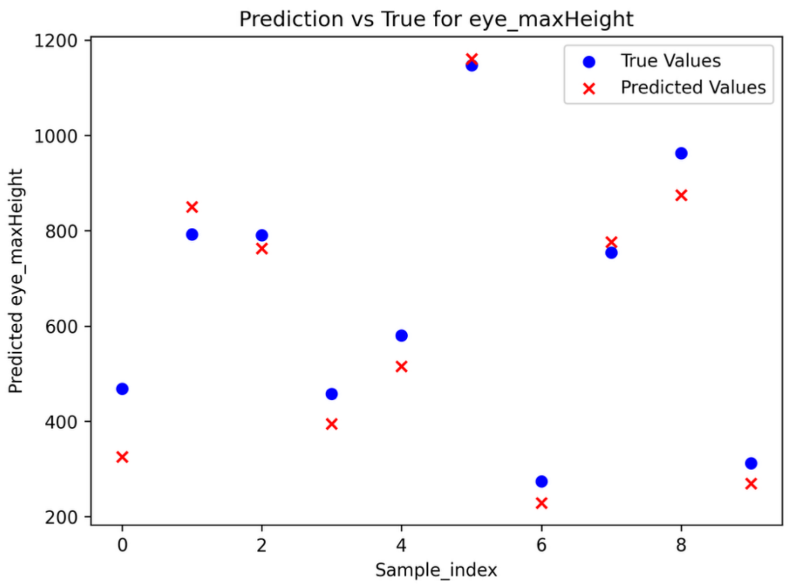


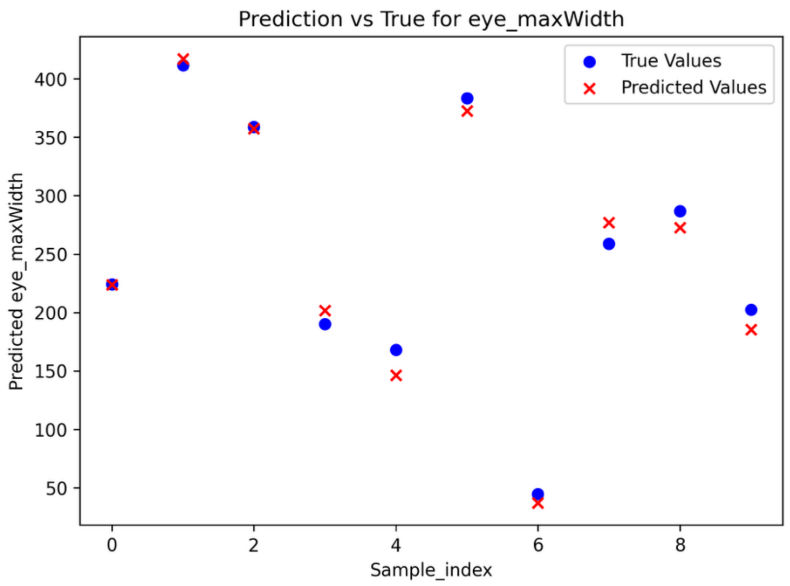


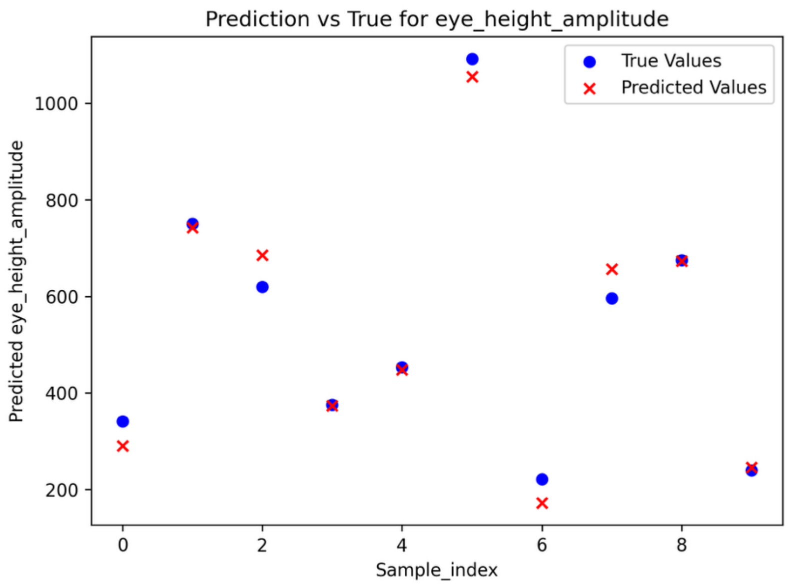
Machine Learning Approaches

Deep Neural Network

- Predicted VS True Values







Conclusions

- Signal Integrity assessment on Silicon Interposers
- X – axis partitioning was implemented
  - 120x speed-up, 97% memory improvement with acceptable error (below 8%)
- Z – axis partitioning was implemented
  - The crosstalk between the metal layers is dominant and the error is high
  - NOT recommended to be integrated to the flow
- Rapid Eye Diagram Metrics Prediction with ML
  - Random Forest Regression (MAPE: 0.26 %)
  - Deep Neural Networks (MAPE: 4.06 %)

Future Work

- Improve the hierarchical partitioning to be applicable in more demanding structures
  - Irregular links, angles, etc.
- Build Algorithms for hierarchical splitting
- Interposer Modeling (Include TSVs and UBM arrays)
- S – parameters prediction with:
  - Graph Neural Networks

Thank You!